

Modeling Anthropogenic CO2 Accumulation: A Bayesian Comparison of Linear and Accelerating Trends

Álvaro Méndez Rodríguez de Tembleque^a

^a *Universita degli Studi di Padova, Department of physics and astronomy “Galileo Galilei”, City, 12345*

Abstract

The Mauna Loa Observatory is a premier baseline site for monitoring historical and current atmospheric carbon dioxide (CO₂) concentrations. This report investigates whether the long-term increase in atmospheric CO₂ is best described by a constant linear increase or an accelerating quadratic trend. Using monthly average CO₂ data from the NOAA Global Monitoring Laboratory, we address missing values and compute annual averages to isolate the long-term trend from seasonal variations. We perform a Bayesian analysis to estimate the parameters of both a linear model and a quadratic model, with a specific focus on the acceleration parameter. The two models are then evaluated and compared using an approximate Bayes factor to determine which model better captures the underlying trend dynamics. Finally, utilizing the best-fitting model, we project the annual atmospheric CO₂ concentrations for the next decade, providing predictive estimates complete with Bayesian credible bands to quantify uncertainty.

1. Introduction

1.1. Scientific Context

1.2. Objectives

2. Data Description and Preprocessing

The dataset used in this analysis consists of monthly average atmospheric CO₂ concentrations measured at the Mauna Loa Observatory, Hawaii, from 1958 to 2026. The data is sourced from the NOAA Global Monitoring Laboratory and is available in a CSV format. Each record includes the date of measurement and the corresponding CO₂ concentration in parts per million (ppm). The dataset contains some missing values, which are we dropped all the rows with missing values limiting our analysis to the period from 1974 to 2026.

The dataset is structured as follows:

- **year:** The year of the measurement.

*Corresponding author

Email address: alvaro.mendezrodriguezdetembleque@studenti.unipd.it (Álvaro Méndez Rodríguez de Tembleque)

- **month**: The month of the measurement.
- **decimal_date**: The decimal representation of the date (e.g., 2020.5 for June 2020).
- **monthly_average**: The average CO2 concentration for that month in ppm.
- **deseasonalized**: The monthly average CO2 concentration with the seasonal component removed.
- **num_days**: The number of days in the month used for averaging.
- **st_dev**: The standard deviation of the daily CO2 measurements for that month.
- **unc_mon_mean**: The uncertainty in the monthly mean CO2 concentration.

2.1. Dataset Overview

See code 1 in Appendix for the implementation.

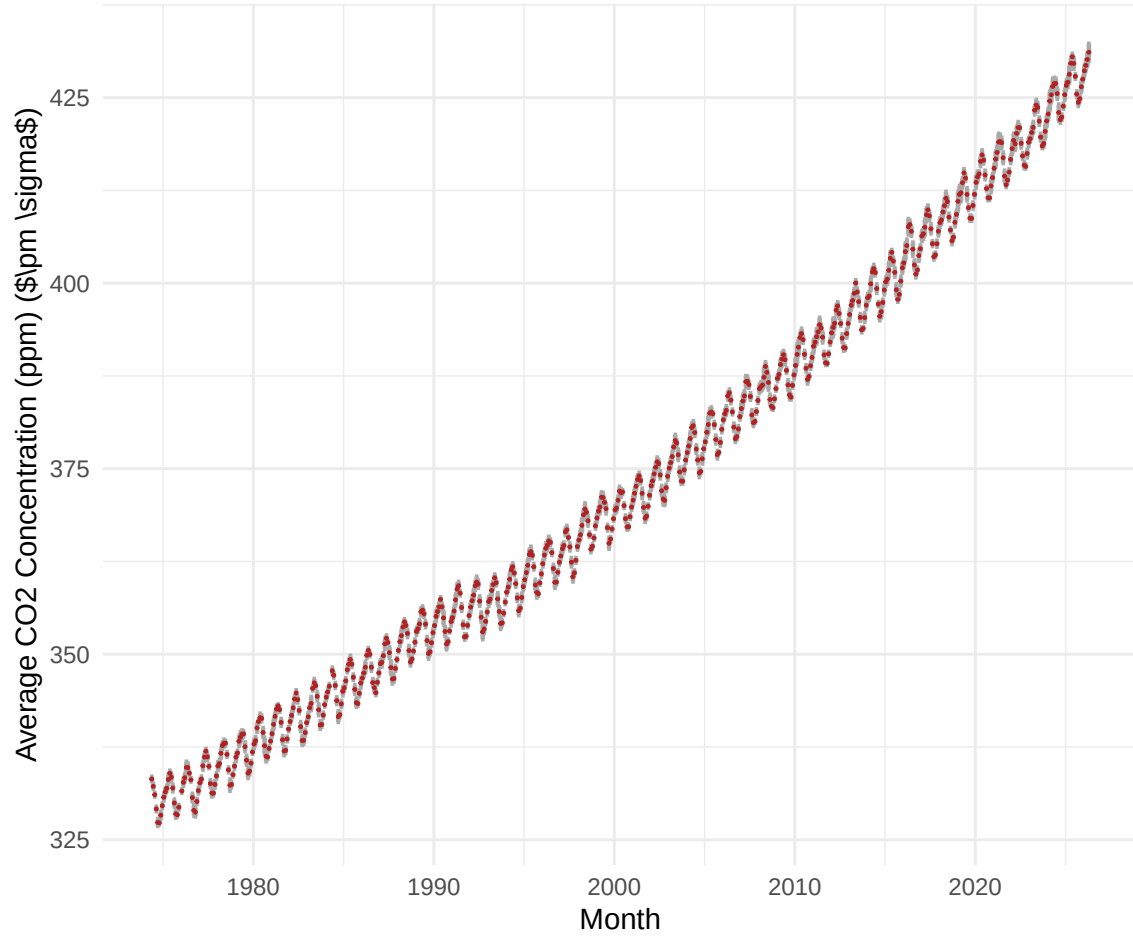


Figure 1: Yearly average CO2 concentrations. The plot shows a clear upward trend in atmospheric CO2 levels over time.

From Figure 1, we observe a clear upward trend in atmospheric CO2 levels over time, with the

data showing a consistent increase in annual average CO2 concentrations. This visual evidence suggests that the long-term trend may not be purely linear, prompting us to consider both linear and accelerating models for further analysis. As we can see due to the incomplete data of the year 2026, the data for 2026 will not be included in the analysis, and our projections will be based on data up to 2025 (the last complete year).

2.2. Data Cleaning

2.3. Aggregation

2.4. Exploratory Visualization

See code 2 in Appendix for the implementation.

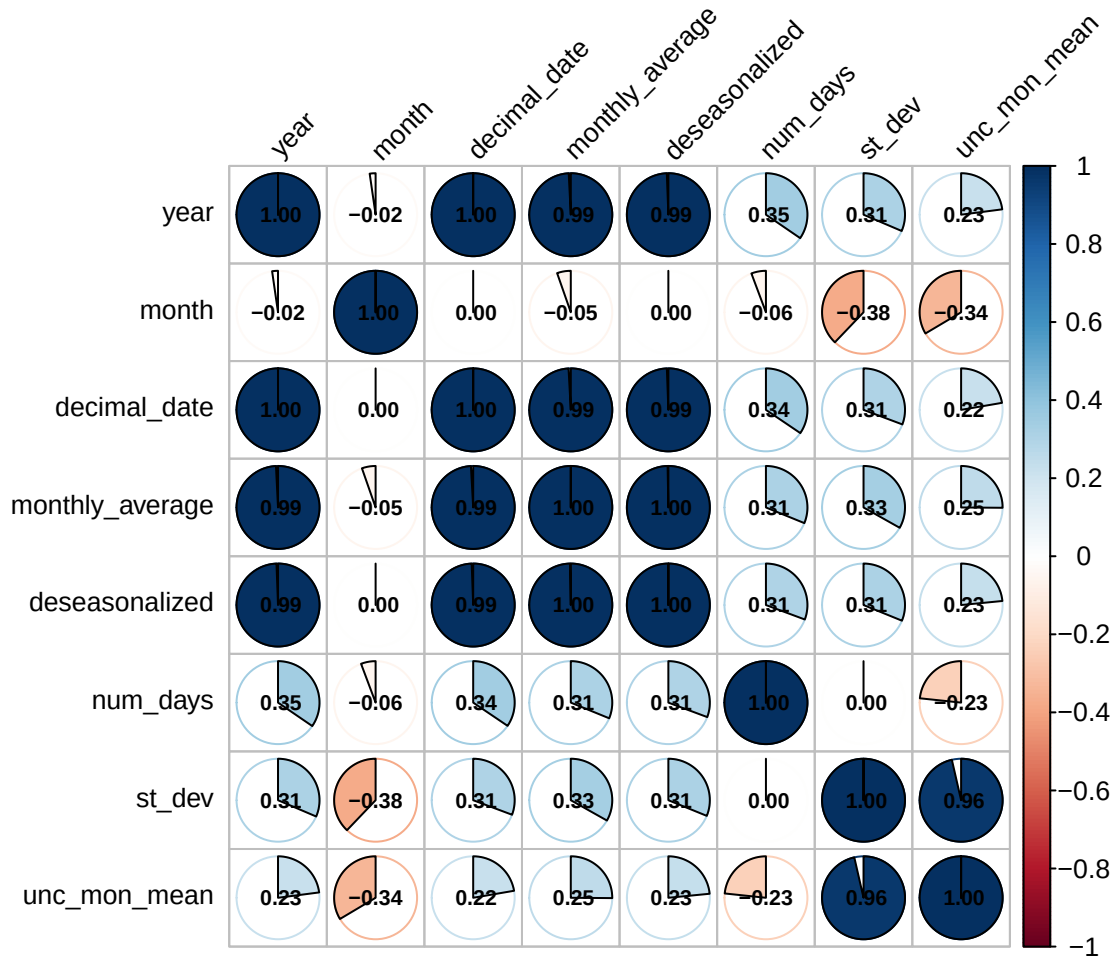


Figure 2: Correlation plot of the variables in the dataset

3. Bayesian Methodology and Models

3.1. Model 1: Linear Trend

3.2. Model 2: Quadratic Trend

3.3. Inference Strategy

4. Results and Inference

4.1. Parameter Estimation

4.2. Model Comparison

5. Predictive Forecasting

5.1. Ten-Year Projections

5.2. Uncertainty Quantification

6. Conclusion

6.1. Summary of Findings

6.2. Limitations and Future Work

6.3. Appendix: Source code

```
1 source("code/Data.R")
2 library(ggplot2)
3
4
5 ggplot(data_no_nan, aes(x = decimal_date, y = monthly_average)) +
6   # Add the error bars first so they sit slightly behind the points
7   geom_errorbar(aes(ymin = monthly_average - st_dev,
8                     ymax = monthly_average + st_dev),
9                 width = 0.2,           # Width of the whisker caps
10                color = "darkgray",    # Color of the bars
11                linewidth = 0.8) +     # Thickness of the bars
12   # Add the scatter points
13   geom_point(color = "firebrick", size = 0.2) +
14   # Clean up the styling
15   theme_minimal() +
16   labs(
17     x = "Month",
18     y = "Average CO2 Concentration (ppm) ($\\pm\\sigma$)"
19   )
```

Listing 1: Plotting the data

```
1 # This is an example of a test file for EDA (Exploratory Data Analysis)
2 source("code/Data.R")
3 library(corrplot)
4
5
6 corrplot(cor(data_no_nan), method = "pie", addCoef.col = "black", tl.cex = 0.8,
7          number.cex = 0.7,
8          tl.col = "black", tl.srt = 45, addrect = 2, rect.col = "blue", rect.
9          lwd = 1.5)
```

Listing 2: Correlating the data

References